

## **BACKLOG BUDDY ? STUDY MATERIAL**

Microprocessors and Microcontrollers ? Unit-wise Lecture Notes

Subject Code: EC402 Semester: 4

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Condensed notes covering all five units: 8086 Architecture, Instruction Set and Assembly Programming, Interrupts and Timers, Peripheral Interfacing, Microcontroller Architecture.

### **Unit 1: 8086 Architecture**

- Definition, key concepts and significance of 8086 Architecture
- Important diagrams / derivations / algorithms for 8086 Architecture
- Solved examples and previous-year questions on 8086 Architecture

### **Unit 2: Instruction Set and Assembly Programming**

- Definition, key concepts and significance of Instruction Set and Assembly Programming
- Important diagrams / derivations / algorithms for Instruction Set and Assembly Programming
- Solved examples and previous-year questions on Instruction Set and Assembly Programming

### **Unit 3: Interrupts and Timers**

- Definition, key concepts and significance of Interrupts and Timers
- Important diagrams / derivations / algorithms for Interrupts and Timers
- Solved examples and previous-year questions on Interrupts and Timers

### **Unit 4: Peripheral Interfacing**

- Definition, key concepts and significance of Peripheral Interfacing
- Important diagrams / derivations / algorithms for Peripheral Interfacing
- Solved examples and previous-year questions on Peripheral Interfacing

### **Unit 5: Microcontroller Architecture**

- Definition, key concepts and significance of Microcontroller Architecture
- Important diagrams / derivations / algorithms for Microcontroller Architecture
- Solved examples and previous-year questions on Microcontroller Architecture

Exam Tips:

- \* Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- \* Solve at least 3 previous year papers under timed conditions before the exam.
- \* Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).

**\* Revise definitions, diagrams and standard derivations ? they earn easy marks.**

\* Practise numerical problems from every unit; papers usually include 2?3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.